



IRB Application

PROPOSED IRB REVIEW TYPE:

Select the type of IRB review requested for your proposed research. The IRB will make the final determination on the type of review for this IRB submission based on the federal research regulations. Click on the options below for more information about the criteria required for each type of IRB review.

- ☒ [Exempt Review](#)
☐ [Expedited Review](#)
☐ [Full Board Review](#)

RESEARCH TITLE

Cognitive Warrant: A Framework and Pilot Methodology for Evaluating Semantic Alignment in Controlled Vocabularies

SECTION A: RESEARCH PERSONNEL

A.1 PRINCIPAL INVESTIGATOR

PRINCIPAL INVESTIGATOR (FACULTY, STAFF, OR STUDENT)

Principal Investigator Name: Nicholas Meister DU ID#: 873740977

Position/Title: ☐ FACULTY ☐ STAFF ☐ GRAD-DOC ☒ GRAD-MASTERS ☐ UNDERGRAD

Department/College/Division: Research Methods and Information Science, Morgridge College of Education

Office/Cell Phone #: 404-731-8015

Email Address: nicholas.meister@du.edu

FACULTY SPONSOR, IF STUDENT INVESTIGATOR

Faculty Sponsor Name: Krystyna K. Matusiak

Credentials/Title: PhD

POSITION: ☒ PROFESSOR ☐ ASSISTANT PROFESSOR ☐ ASSOCIATE PROFESSOR ☐ STAFF ☐ OTHER:

Department/College/Division: Research Methods and Information Science, Morgridge College of Education

Office/Cell Phone #: 303-871-6163 (Office)

Email Address: Krystyna.Matusiak@du.edu

Student Project: ☐ THESIS ☐ DISSERTATION ☒ CAPSTONE ☐ OTHER: EXPLAIN BELOW

If 'OTHER' explain here: [CLICK HERE TO ENTER TEXT](#)

A.2 DU RESEARCH PERSONNEL

NEW REQUIREMENT: All research team members must create an IRBNet account and link their CITI training account to it (Do not upload CITI certificates for DU personnel). Please follow these instructions to register an account in IRBNet, link your CITI account to IRBNet, and share project access for all team members and link training:

- [Register an Account in IRBNet](#)
- [Linking Your CITI Account to IRBNet](#)
- [Sharing Access to Your Projects in IRBNet](#)

All research personnel are required to complete & maintain valid human subjects protection training through the CITI program. IRB-approved courses include Human Subjects Research - Social Behavioral Educational Research (SBER) or Biomedical Research. Certificates are valid for four (4) years after the date of completion.

Please note: Responsible Conduct of Research (RCR), Conflict of Interest (COI), and Good Clinical Practice (GCP) courses are separate from the human subjects research course requirement and may not be used as a substitute.

List all personnel, including yourself as the PI and your faculty sponsor if you're a student, who are engaged with DU and will have interaction or intervention with participants or have access to any identifiable, private data, or biospecimens during the research (e.g., administering consent, collecting data, conducting interviews or tests, analyzing identifiable data, etc.). Video and audio recordings are considered identifiable data.

TABLE A.2.1

List all personnel, including yourself as the PI and your faculty sponsor if you're a student.

NAME	DEGREE(S)	DEPARTMENT	POSITION/TITLE	RESEARCH ROLE
Nicholas Meister	MS, MA	Research Methods and Information Science	Student	PI
Krystyna Matusiak	PhD	Research Methods and Information Science	Faculty	Faculty sponsor

To add additional personnel, click on the + at the end of each row.

A.3 UNAFFILIATED INVESTIGATORS AND INDIVIDUALS

A.3.1 Will the proposed research project involve other investigators who are not affiliated with DU, or individuals from an external organization or institution? ☐ Yes ☒ No

If **NO**, continue to [Section B: Funding and Financial Conflict of Interest](#)

If **YES**, list all unaffiliated investigators and individuals in [TABLE 3.1](#) who will be directly involved in interacting with potential participants (e.g., administering consent, conducting interviews, etc.) or viewing identifiable data.



Individuals only involved in distributing recruitment materials, on behalf of the PI, and will not consent or participate in any study activities, should not be listed as unaffiliated individuals below in section A.3.2.

A.3.2 Has a request been issued for an external institution or organization to serve as the IRB of Record for this project and enter into a reliance agreement? ☐ Yes ☐ No

If **YES**, **STOP. Do not complete this form.** Instead, complete [Appendix Q: Ceding to an External IRB](#) and follow the instructions for requesting DU to cede to an external IRB to serve as the IRB of Record for a collaborative project (does NOT apply to Exempt research or international institutions; continue with this form).

If **NO**, you will need to complete **Section L: Multi-Site/Collaborative Research with External Institutions or Organizations & Unaffiliated Investigators** of this application.

TABLE 3.1 – Unaffiliated Investigators/Individuals

To add non-affiliated individuals, click on the + at the end of each box.

NOTE: All unaffiliated investigators/individuals added to this research project are required to complete & maintain valid human subjects protection training through the Collaborative Institutional Training Initiative (CITI) Program or equivalent training offered by their institution.

SECTION B: FUNDING AND CONFLICT OF INTEREST (COI)

B.1 FUNDING

B.1.1 Will the proposed research be funded? ☐ Yes ☒ No (if “no”, proceed to **B.2 Conflict of Interest**)

If **YES**, please select the funding source(s) for the proposed research:

- ☐ Federally-funded (i.e., NIH, NSF, DoD, etc.): (provide name of funder: [CLICK HERE TO ENTER TEXT](#))
- ☐ State or Local Government Grant: (provide name of funder: [CLICK HERE TO ENTER TEXT](#))
- ☐ Contract (i.e., private industry) (provide name of funder: [CLICK HERE TO ENTER TEXT](#))
- ☐ DU Internal Awards (i.e., PinS, Summer Research Grant, etc.)
 - ☐ Faculty Research Fund
 - ☐ PROF
 - ☐ PinS
 - ☐ Summer Research Grant
 - ☐ Other: [CLICK HERE TO ENTER TEXT](#)
- ☐ Private Foundation (provide name of funder: [CLICK HERE TO ENTER TEXT](#))
- ☐ Scholarship/Fellowship (provide name of funder: [CLICK HERE TO ENTER TEXT](#))
- ☐ Gift (provide name of funder: [CLICK HERE TO ENTER TEXT](#))
- ☐ Other: [CLICK HERE TO ENTER TEXT](#)

B.1.2 IF FUNDED PROJECT:

Grant Proposal Title: [CLICK HERE TO ENTER TEXT](#)

PI on Grant: [CLICK HERE TO ENTER TEXT](#)



If this project is externally-funded, you must have a FCOI Disclosure on file in InfoEd.

B.2 CONFLICT OF INTEREST (COI)

The term “conflict of interest (COI) in research” refers to situations in which financial or other personal considerations may compromise — or have the appearance of compromising — an investigator’s professional judgment in conducting or reporting research. A COI depends on the situation and not on the actions or character of an individual investigator.

It is important that researchers involved in human research do not have or appear to have a COI—including a financial interest—related to any of the studies in which they participate.

Federal regulations, state laws and University policies require that faculty members submit financial disclosure forms at the time that a proposal is submitted for funding.

For more information regarding COI in research, please refer to our [website](#).

B.2.1 If this project is externally funded, has the Principal Investigator and any other DU individuals responsible for the design, conduct, or reporting of research completed the [Research Financial Conflict of Interest Disclosure Form](#) within the last 12 months? ☐ Yes ☐ No ☒ N/A

If **NO**, please proceed to [InfoEd](#) to complete this disclosure.

B.2.2 Does the DU Principal Investigator or any of the individuals listed on the research protocol have any financial or non-financial conflicts of interest that reasonably appears to be related to the research?

Examples of non-financial conflicts of interest include things such as personal and professional relationships, affiliations, etc. ☐ Yes ☒ No

If **NO**, there are no individuals who have potential COI related to this Research Project. Proceed to [B.2.3](#)

If **YES**, please describe the nature of the COI.

[CLICK HERE TO ENTER TEXT](#)

B.2.3 If this research is or may be federally-funded, has the Principal Investigator and all of individuals responsible for the design, conduct, or reporting of the research completed the Research Conflict of Interest training curriculum through the CITI Program? ☐ Yes ☐ No ☒ N/A

If **YES**, proceed to [Section C: Research Purpose](#)

If **NO**, this training must be completed through the CITI Program curriculum affiliated with DU at: citiprogram.org prior to the expenditure of federal funds and issuing IRB approval.



SECTION C: RESEARCH PURPOSE

C.1 PROJECT SUMMARY

Provide a **brief, concise summary** of the purpose of your research project.

This study investigates whether the categorical and hierarchical distinctions encoded in the Library of Congress Subject Headings (LCSH) correspond to measurable patterns in human semantic cognition — a property the investigator terms "cognitive warrant." Using a stratified sample of 50 lexically ambiguous word senses drawn from a validated psycholinguistic dataset (Trott & Bergen, 2023), the study maps each word sense to an authorized LCSH heading, computes structural distances between headings using graph-theoretic methods applied to the LCSH hierarchy, and estimates distributional semantic distances using BERT sentence embeddings. The current scope of the study comprises three components: (1) proof-of-concept analysis using Trott and Bergen's publicly available stimuli and data to validate key independent variables and the analysis pipeline; (2) expert consultant validation of the PI's LCSH stimulus mappings by two DU librarians with cataloging experience, using an inter-rater reliability protocol (Cohen's kappa ≥ 0.80 threshold) with a structured consensus meeting; and (3) pilot testing of two online behavioral tasks — a response-time sensibility judgment paradigm and a subject heading appropriateness rating task — recruiting 20 participants (with a minimum reporting threshold of 15 participants) through the DU Psychology Department's Research Participation System (SONA). The pilot study establishes the theoretical framework, validates the methodology and technical infrastructure, and collects preliminary behavioral data in preparation for a fully powered, preregistered follow-on study.

C.2 PROJECT RESULTS

C.2.1 Specify what you plan to do with the results of your research

- ☐ Publish in a professional journal
- ☐ Poster Presentation
- ☐ Conference Presentation
- ☒ Fulfill a dissertation, thesis or capstone project requirement
- ☐ Validate a new intervention, product, or test
- ☐ Other: [CLICK HERE TO ENTER TEXT](#)

C.3 LOCATION OF RESEARCH

PLEASE NOTE: Students engaging in research abroad should register at abroad.du.edu. Faculty and staff are registered by virtue of their required booking in Concur. For more information about DU's [international travel policy](#), please visit: <https://www.du.edu/risk/international-travel>

C.3.1 List the site(s) or location(s) where the research team will conduct the research (e.g., DU campus locations, clinic, school, community group/agency, village, lab, etc.)

The review of stimuli will be completed by two DU librarians remotely on their own computers. The consensus meeting with both librarians will be conducted in the Anderson Academic Commons. The pilot testing of the



two surveys, with participants recruited through the DU Psychology Department's Research Participation System (SONA), will be conducted online through the Pavlovia platform.

C.3.2 If the research will physically be conducted outside of the United States, please complete and submit: [Appendix G: International Research](#).

SECTION D: RESEARCH PROCEDURES

D.1 CATEGORIES OF RESEARCH

The research will involve the following (check all that apply):

- | | |
|--|---|
| ☐ Activity Observations | ☐ Ethnographic |
| ☒ Analysis of Existing Data/ Secondary Data | ☐ Field Research (Complete Appendix B) |
| ☐ Analysis of Biological Specimens | ☐ Evaluation of social or educational programs |
| ☐ Audio/Video Recording | ☐ Focus Groups |
| ☐ Classroom Observations | ☐ International Research |
| ☐ Clinical Procedures | ☐ Interviews |
| ☐ venipuncture/finger stick | ☐ Investigational Devices |
| ☐ blood pressure | ☐ Investigational Drugs/Biologics |
| ☐ saliva or hair sampling | ☐ Journaling/Diaries |
| ☐ other: CLICK HERE TO ENTER TEXT | ☐ Medical Chart/Records Review (HIPAA may apply) |
| ☐ Collection/Handling of Biospecimens - any protocol involving handling of human tissue/fluids (not including hair or saliva) must obtain Institutional Biosafety Committee (IBC) approval PRIOR TO obtaining IRB approval. The IBC approval letter must be attached to the IRB application in IRBNet. | ☐ Observation of online activity (Complete Section D.3) |
| ☒ Cognitive & Perceptual Experiments | ☒ Online Surveys (Specify Platform(s)): Qualtrics, Pavlovia |
| ☐ Education Records | ☐ Photography |
| ☐ EKG/EEG/fMRI/Chest X-Ray | ☐ Questionnaires/Surveys |
| ☐ Epidemiological Studies | ☐ Use of previously collected tissues (e.g. BioBank) |
| | ☒ Other: Consultation with DU librarians about experimental stimuli |

D.2 STEP-BY-STEP DESCRIPTION OF RESEARCH PROCEDURES

Provide a step-by-step description of what each participant will be asked to do, the duration, and state the where, when, and by whom the research will be conducted (include all procedures checked above. DO NOT include recruitment or consent information here).

Please see the Supplemental Procedures Document submitted with this application package for a step-by-step description of what each participant will be asked to do as well as what each DU librarian (acting as expert consultant and not participant) will be asked to do. See Appendix O for details about the analysis of existing data. The following is a **very brief summary**. Two DU librarians will independently evaluate the PI's LCSH stimulus mappings using an online Qualtrics interface (~90–120 minutes, asynchronous, self-paced). They will



then attend a single, structured joint consensus meeting with the PI to resolve mapping discrepancies (~60–90 minutes, in person at Anderson Academic Commons or via Zoom). Total time commitment per librarian: approximately 2.5–3.5 hours. Conducted by Nicholas Meister (PI), supervised by Dr. Krystyna Matusiak. Location: remote (Task 1); DU campus or Zoom (Task 2). Anticipated timing: May–June 2026. SONA pilot testing participants: Participants will complete two online computerized tasks hosted on Pavlovia, followed by a brief demographic questionnaire, and an anonymous feedback survey, remotely on their own computers. (1) Sensibility judgment task: participants read short sentences and press F or J to indicate whether each sentence makes sense grammatically (~10 minutes). (2) LCSH appropriateness rating task: participants read sentence pairs and rate the appropriateness of proposed Library of Congress Subject Headings on a 1–7 scale (~28 minutes). (3) Brief demographic questionnaire (~1 minute). (4) Anonymous qualitative feedback survey (~4 minutes). Task order is counterbalanced between participants with a brief break between surveys. Total session duration: approximately 45–55 minutes. Conducted by Nicholas Meister (PI), supervised by Dr. Krystyna Matusiak. Location: online via Pavlovia (remote). Anticipated timing: May–June 2026. Analysis of existing data (Trott & Bergen, 2023). No participant involvement. Please see Appendix O.

D.3 OBSERVING INTERNET ACTIVITY

Will you be observing internet activity (e.g., chatrooms, social media, web browsing)?

- ☐ Yes
☒ No (**proceed to Section E**)

If Yes, describe the setting and nature of the online activity being observed:

[CLICK HERE TO ENTER TEXT.](#)

If Yes, does the privacy policy for the site where you will be observing activity include reference to research or other information relevant to your research? Please explain.

[CLICK HERE TO ENTER TEXT.](#)

If Yes, will subjects be aware that their activity is being observed (consider chat room access, privacy setting on social media sites, privacy warnings on web pages, etc.)?

- ☐ YES
☐ NO

SECTION E: RECRUITMENT - PARTICIPANT AND POPULATION SELECTION

In determining whether selection of subjects for this project is fair and equitable, the IRB will consider the purpose of the research, the setting in which the research will be conducted as well as additional safeguards to protect vulnerable populations, such as children, prisoners, pregnant women, individuals with diminished decision-making capacity, or economically or educationally disadvantaged persons.



E.1 RESOURCES FOR IDENTIFYING AND RECRUITING PARTICIPANTS

E.1.1 Identify the resources or individual contacts that will be utilized to identify eligible participants. (check all that apply):

- ☐ Requesting potential contacts from outside source (may require Data Use Agreement)
- ☐ Requesting student email addresses from a DU department or Registrar's Office
- ☐ Requesting others to send recruitment email on PI's behalf
- ☐ Snowball sampling
- ☐ Referrals from professional contacts/organizations
- ☒ Participant pools (i.e., SONA)
- ☐ Crowdsourcing (Specify Platform(s)): [CLICK HERE TO ENTER TEXT](#)
- ☐ IRB-approved screening and/or recruitment protocol and/or recruitment database
- ☐ Review of publicly available records
- ☐ Recruitment in classrooms
- ☐ Recruiting researchers' students or staff
- ☐ Review of student records (requires FERPA authorization)
- ☐ Review of medical or counseling/therapy records (requires HIPAA authorization)
- ☐ Recruiting individuals from the European Union (EU)
- ☐ Other: [CLICK HERE TO ENTER TEXT](#)

E.2 TARGETED PARTICIPANTS AND POPULATIONS

E.2.1 Select the following participants and populations that will be targeted for this research (check all that apply):

- ☒ Healthy Participants
- ☐ Veterans
- ☐ Elderly
- ☐ Employees, Specify Employer: [CLICK HERE TO ENTER TEXT](#)
- ☐ Individuals from the European Union (EU)
- ☐ Children or Minors, (complete [Appendix F](#))
- ☐ Non-English Speaking Subjects (complete [Appendix K](#))
- ☐ Pregnant Women, Fetuses, Neonates (complete [Appendix H](#))
- ☐ Adolescents detained in juvenile detention facility (complete [Appendix D](#))
- ☐ Prisoners (complete [Appendix D](#))
- ☐ Students or Teachers (Pre-school, K-12 public/private school, university) (complete [Appendix M](#))
- ☐ Adults with Diminished Decision-Making Capacity (complete [Appendix C](#))
- ☐ Institutionalized Individuals (not prisoners) (complete [Appendix C](#))
- ☐ Undocumented Immigrants (complete [Appendix C](#))
- ☐ Indigenous Peoples (complete [Appendix C](#))
- ☐ Educationally Disadvantaged Individuals (complete [Appendix C](#))
- ☐ Low-income and/or Homeless Individuals (complete [Appendix C](#))
- ☐ None Apply

If any vulnerable individuals or target populations were identified in [E.2.1](#), you are required to document additional protections using the associated appendices.

[CLICK HERE TO ENTER TEXT](#)

E.3 INCLUSION AND EXCLUSION CRITERIA

Research that has the potential to benefit a variety of individuals should target a population of subjects that is diverse enough to distinguish differing effects, risks, and benefits. No group or individual subject should be excluded without a scientifically sound reason or a requirement of the regulations to do so.

E.3.1 Describe the criteria that define who will be included or excluded in your proposed study sample.

Indicate specifically whether you will include or exclude any special populations. ***You may not include members identified in [section E.2](#) as subjects in your research unless you indicate this in your inclusion criteria.***

Inclusion Criteria: Inclusion criteria are defined as the key features of the target population that the investigators will use to answer their research question. Typical inclusion criteria include demographic, clinical, and geographic characteristics.	Age: 18 years or older. Language: Native English speakers or native-level English proficiency. Technical Access: Ability to access and complete an online study via computer or laptop with an internet connection and keyboard. Reading Ability: Ability to read English text at a normal reading level without significant visual or reading impairments.
Exclusion Criteria: Exclusion criteria are defined as features of the potential study participants who meet the inclusion criteria but present with additional characteristics that could interfere with the success of the study or increase their risk for an unfavorable outcome.	Visual/Reading Impairments: Self-reported severe visual impairments or reading disabilities that would prevent accurate reading of sentence stimuli.

E.3.2 Describe the process for assessing and documenting a participant's qualifications based on the inclusion and exclusion criteria identified in [E.4.1](#) above.

Participants will access the study through the DU Department of Psychology Research Participation System (SONA). The postings will clearly state inclusion criteria (age 18+, native English speaker, must use a computer/laptop with a keyboard). Before viewing the informed consent form, participants will complete a brief eligibility screening with the following questions: "Are you 18 years of age or older?", "Are you a native English speaker or do you have native-level English proficiency?", "Are you completing this study on a computer or laptop with a keyboard?", and "Can you read English text comfortably without significant visual or reading difficulties?" Participants must answer "Yes" to questions 1-4 to proceed to the consent form. No personally identifiable information will be collected from ineligible participants. At the end of the surveys, participants will complete a brief demographic questionnaire that includes: age, gender, native-level English



speaker status, and familiarity with library subject headings. All screening responses, consent records, and demographic data will be stored securely in the SONA platform database. Participants will be assigned anonymous participant IDs, with no linking of SONA identities to experimental data.

E.4 ENROLLMENT OF PARTICIPANTS

E.4.1 Indicate the total number of subjects to be recruited/enrolled in this research.

The pilot test will enroll 20 participants recruited through the DU Psychology Department's Research Participation System (SONA), with a minimum reporting threshold of 15 participants. The originally planned full experiments will be conducted under a subsequent IRB application.

E.4.2 If this study will be conducted in multiple locations, indicate the total number of subjects to be enrolled in each participating site.

N/A

SECTION F: PARTICIPANT RECRUITMENT PROCEDURES

F.1 RECRUITMENT PLAN

F.1.1 Identify where the research team will identify and recruit potential participants. If specific individuals or organizations will be used for recruiting purposes only, explain how information will be provided to them and how potential participants will contact the researcher.

Participants will be recruited from the DU Psychology Department's participant pool on the Sona Systems platform.

F.1.2 Will you be screening for eligibility before consenting potential subjects into the study?

- ☐ No. Proceed to section F.2.
- ☒ Yes. Complete the information below.

Indicate what you will use to screen potential subjects:

- ☒ Survey/Questionnaire
- ☐ Other: [CLICK HERE TO ENTER TEXT](#)

The screening instrument must include an introduction and the following statements (edit appropriately):

- You will be notified if you are eligible for this study. The information you provide below **will/will not** be used in the study.
- If you are not eligible for this study, you **will/will not** be notified, and any information you provide below will be destroyed.

You do not need to obtain informed consent to screen for eligibility. Potential subjects should not be asked to consent to participating if they are not eligible for the study in the first place.



Screening instruments should only include items or questions based on the inclusion and exclusion criteria listed in item E.4.1, unless you can provide justification for additional items.

Provide justification for additional screening items:

N/A

F.2 RECRUITMENT MATERIALS

F.2.1 Check all recruitment materials that will be used in the research. Copies of all recruitment materials must be included in the IRBNet package for IRB approval prior to distribution and use (check all that apply):

- ☐ Advertisements/Public Service Announcements
- ☐ Emails for recruitment - if selected, describe how email addresses will be obtained: [CLICK HERE TO ENTER TEXT](#)
- ☐ Recruitment Flyer (hard copy and/or electronic)
- ☐ Letters or information sheets for participants
- ☐ Scripts or guides that will be used for in-person or telephone recruitment interviews
- ☐ Social Media Postings
- ☐ Printouts of web postings or pages used for direct recruitment

☒ Other – describe: This study will be posted on the DU Department of Psychology Research Participation System (SONA). The postings will describe the two online surveys and clearly state inclusion criteria (age 18+, native English speaker, must use a computer/laptop with a keyboard).

Note: Please ensure items selected above are all uploaded in your IRBNet package for review.

F.3 OBTAINING APPROVAL FROM EXTERNAL SITES

F.3.1 Describe any approvals that will be obtained prior to commencing the research (e.g., school, external site, DU biosafety committee, community leader, Dept of Corrections, etc.) as identified in [C.3 – Location of Research](#).

N/A

If research is conducted at an off-campus site (i.e., community center, school, etc.) or if recruiting potential study participants from that site (i.e., employees, students, etc.) a signed site permission/letter of support (a template is available on the DU IRB Forms webpage) must be obtained from the external site confirming their support of the proposed research and their approval to have the research be conducted at their site.

All letters of support from external sites must be signed and include the title of the signing party posted in the IRBNet project submission. Letters of support must be obtained and reviewed before final IRB approval will be issued.

F.4 OBTAINING RECORDS AND DATA

F.4.1 Will participants be recruited by obtaining records (e.g., educational records, medical or counseling/therapy records, databases, etc.)? ☐ Yes ☒ No

If **No** proceed to [F.5 – Subject Compensation](#)



If **Yes**

- a. Will this include paper files? ☐ Yes ☐ No

PLEASE DESCRIBE: [CLICK HERE TO ENTER TEXT](#)

- b. Will this include electronic files? ☐ Yes ☐ No

PLEASE DESCRIBE: [CLICK HERE TO ENTER TEXT](#)

F.4.2 Will **student educational records** be requested or reviewed for research purposes? ☐ Yes ☐ No

If **NO**, proceed to [F.4.3](#)

If **YES**, obtaining **Family Educational Rights and Privacy Act (FERPA) authorization is required**. FERPA authorization language should be included in the consent (refer to [Informed Consent Template](#) for appropriate language) and the [Student Records Request Form](#) must be completed and sent to the Office of Institutional Research. For further guidance on FERPA regulations and procedures, please see the [FERPA Guidance](#) document.

NOTE: *If a researcher has access to student records as part of their professional or academic position, they are not allowed to access those records for research purposes without obtaining proper written authorization from the student.*

F.4.3 Will an **individuals' medical or counseling/therapy records** be reviewed or requested from a covered entity for research or recruitment purposes? ☐ Yes ☐ No

If **NO**, proceed to [F.4.4](#)

If **YES**, obtaining **Health Insurance Portability and Accountability Act (HIPAA) authorization is required**. HIPAA authorization must be provided by the healthcare provider or individual patient or client. Refer to the [Informed Consent Template](#) for appropriate language) or be obtained as a separate document.

NOTE: *If a researcher has access to an individuals' medical or counseling/therapy records as part of their professional or clinical position, they are not allowed to access those records for research purposes without obtaining proper authorization from the individual.*

F.4.4 Will **identifiable data** be requested and provided by an outside entity for research purposes? ☐ Yes ☐ No

If **NO**, proceed to [F.5 – Subject Compensation](#)

If **YES**, complete the Office of Intellectual Property and Technology Transfer (OIPTT) intake form [here](#). The DUA process and review is conducted separately from the IRB. After the DUA form and agreement has been reviewed and approved, a copy will be issued to the PI. The PI is responsible for attaching the fully executed DUA document to their approved IRBNet project as a new package labeled "Other."

If **de-identifiable data** is requested from an outside entity, without any codes or access to the key to the codes, it is at the discretion of the outside entity whether they will require a DUA.



F.5 SUBJECT COMPENSATION

F.5.1 Will subjects be offered compensation for participating in the research? ☒ Yes ☐ No

If **NO**, proceed to [Section G: Consent Procedures](#)

If **YES**, check all that apply:

- ☐ Gift card (specify type): [CLICK HERE TO ENTER TEXT](#)
- ☐ Monetary cash payment
- ☐ Check
- ☒ Class credit (If selected, complete [F.5.4](#) below)
- ☒ SONA chit
- ☐ Gift
- ☐ Drawing/Lottery for gift card or cash
- ☐ Crowdsourcing platform
- ☐ Other: [CLICK HERE TO ENTER TEXT](#)

F.5.2 Describe the amount and the schedule for issuing payments as well as any conditions for participant receiving the compensation (e.g., prorating payment, timing of the drawing or payment). If payment by check, gift card, or cash will be used, please refer to the [Sponsored Program Participant Payments](#).

Participants from undergraduate psychology courses will earn credit towards their course.

F.5.3 Will an individual participant receive more than \$600 per calendar year for participating in this research project? ☐ Yes ☒ No

If **YES**, additional information must be included in the consent document regarding the IRS reporting requirement and 1099 language (refer to the [Informed Consent Template](#) for specific language).

F.5.4 If students will receive extra credit or course credit (i.e., SONA chits), state the alternative method(s) that will be available for earning credit for those who do not wish to participate. The alternative method for earning extra credit must be provided in the recruitment materials and included in the consent or information sheet.

Participants will be clearly informed that 1) their participation is entirely voluntary, 2) they are contributing to basic scientific research, 3) they will not receive direct personal benefit beyond research experience and course credit, 4) they may withdraw at any time without penalty, and 5) alternative methods of earning research credit are available through other studies listed on the DU Department of Psychology Research Participation System (SONA).

SECTION G: CONSENT PROCEDURES

G.1 CONSENT OPTIONS

Select the consent document(s) that will be utilized in your research (please be sure to only select the option from the review category for your research project). All consent templates are located on the [IRB Forms webpage](#).



Options for Exempt Review Projects

- ☐ [Exempt Research Information Sheet](#) – Complete and upload to IRBNet package for review. You will need to offer a copy to the participants.
- ☒ [Consent for Online Surveys](#) – Consent language to be used with online surveys or questionnaires. Complete and upload to IRBNet for review.
- ☐ **Non-English Reading/Speaking Participants** - Complete the [Exempt Research Information Sheet](#) in both English and the translated language. Upload to IRBNet for review along with [Appendix K](#).

Options for Expedited & Full Board Review Projects

- ☐ **Written Informed Consent** (physical, hand-written signature) - (complete [Expedited/Full Board Consent template](#)). A hard copy of the consent form must be provided unless providing a copy increases potential risk to the participant.
- ☐ **Electronic Consent** - no written hardcopy signature, but consent obtained and completed using an authenticated secure signature platform (i.e., DocuSign or REDCap). **Note: Qualtrics may not be used to obtain electronic signatures.** (Complete [Expedited/Full Board Consent template](#) and remove signature/date line). **Note:** Research participants must be offered a hard copy or electronic version of the consent form they can retain for their records.
- ☐ **Waiver of Written Documentation**, no written or electronic signature (complete [Appendix A](#) and [Expedited/Full Board Consent template](#) and remove signature/date line). Note: Research participants must be offered a hard copy or electronic version of the consent form they can retain for their records.
- ☐ **Verbal Consent – Waiver of Written Documentation** (complete [Appendix A](#) and [Verbal Consent Form](#)). Provide a copy of the Verbal Consent Form. If appropriate, a set of questions may be included to test their comprehension.
- ☐ **Translated Consent Form (Non-English language)**, (complete [Appendix K](#) and the [Informed Consent Document template](#) formatted in the English language). Provide a copy of the fully Translated **Short Form Consent** in Non-English language (to the participant) unless providing a document increases a potential risk to the participant. If appropriate, a set of questions may be included to test their comprehension.
- ☐ **Verbal Consent (Non-English language)**, (complete [Appendix K](#) and [Verbal Consent Script template](#) formatted in English language). Provide a copy of the Short Form Consent in Non-English language unless providing a document increases a potential risk to the participant. If appropriate, a set of questions may be included to test their comprehension.
- ☐ **Short Form Consent**, required if consenting non-English speaking potential participants and an interpreter is used. Complete the [Short Form Consent template](#). Provide a copy of the Short Form Consent in the English-version for IRB review. Provide a copy of the non-English version to the participant unless providing a document



increases a potential risk to the participant. If appropriate, a set of questions may be included to test their comprehension.

- ☐ **Full Waiver or Alteration of Consent Elements** - (complete [Appendix A](#), and if alteration of consent requested, complete [Informed Consent Document template](#) as appropriate). An alteration of consent is requested if any of the required elements of an informed consent are altered or removed.
- ☐ **Written Assent Document** (aged 7-12 year old participants) - (complete [Assent Document template](#))
- ☐ **Written Assent Document** (aged 13-17 year old participants) - (complete [Assent Document template](#))
- ☐ **Waiver of Assent** (newborn-6 year old participants) – (complete [Appendix A](#) and [Parental Permission Form template](#))
- ☐ **Written Parental Permission Form** (hardcopy signature obtained) – (complete [Parental Permission Document template](#)) Parents or guardians must be provided with a hard copy or electronic version of the parental permission form they can retain for their records.
- ☐ **Parental Permission Form – Electronic Documentation** – no written hardcopy signature but permission is obtained and completed on a secure electronic signature platform (i.e., DocuSign or REDCap. **Note:** Qualtrics may not be used to obtain electronic signatures.) **Note:** Parents or guardians must be provided with a hard copy or electronic version of the consent form that they can retain for their records. (complete [Parental Permission Document template](#) and provide copy of Investigator Letter to be distributed to parent/guardian)
- ☐ **Full Waiver of Parental Permission Form** – (complete [Appendix A](#)) , this type of waiver may only be granted by the IRB for minimal risk studies.

G.2 OBTAINING INFORMED CONSENT, ASSENT, OR PARENTAL PERMISSION

G.2.1 Specify where the informed consent, assent, and/or parental permission process will take place.

Consent for online surveys will take place online via the SONA platform (participants will complete experiments remotely on their own computers).

G.2.2 Specify who on the research team will administer the consent, assent or parental permission.

Nicholas Meister (PI, MLIS student)

G.2.3 If applicable, describe how hard copies or written consent electronically, assents or parental permission forms will be collected, stored and managed. If research will be conducted outside or away from the DU campus, explain how the documents will be securely transferred back to the secure storage site.

All screening responses and consent records will be stored securely in the SONA platform database.

Participants will be assigned anonymous participant IDs, with no linking of SONA identities to experimental data. No personally identifiable information will be collected from ineligible participants.

G.2.4 If obtaining written consent electronically, indicate which platform will be used (Qualtrics may not be used):



- ☐ Docusign (Faculty: check with your department for availability. Students may request an account [here](#))
☐ REDCap

G.3 USING TRANSLATORS AND INTERPRETERS

G.3.1 Will this research project enroll non-English speaking persons? ☐ Yes ☒ No

If **NO**, proceed to [G.4](#)

If **YES**, complete [Appendix C – Populations Requiring Additional Safeguards](#)

G.3.2 Will this research project use translators and/or interpreters to communicate with non-English speaking persons? ☐ Yes ☐ No

If **NO**, proceed to [G.4](#)

If **YES**, complete [Appendix K: Certificate of Translation & Use of Interpreters](#)

G.4 VERIFYING COMPREHENSION AND DOCUMENTATION OF CONSENT

G.4.1 Describe how comprehension about the research will be verified.

The Consent for Online Surveys will include the following information to promote comprehension: what participants will do (read sentences, make grammatical judgments, rate subject headings), time commitment (45–55 minutes total), voluntary nature of participation, right to withdraw at any time, and confidentiality of information (SONA separately stores their Identity Code while responses are sent to a different database for retrieval by the researcher). Subjects will be required to scroll through the entire Consent for Online Surveys before the "I agree to participate" button becomes active. Before beginning each survey, participants will view detailed instruction screens explaining: the task they will perform, how to respond (keyboard keys for Survey 1, rating selections for Survey 2), that they should respond accurately (and, for Survey 1, as quickly as possible), and that there are no right or wrong answers for the rating task. Instructions will include visual examples of the task interface. Participants must click "I understand" to proceed, indicating active engagement with the instructions. In Survey 1, participants complete four practice trials before the main experiment. Immediate feedback is provided after each practice trial, indicating whether their response was correct or incorrect. This allows participants to demonstrate understanding of the task and correct any misunderstandings before critical trials begin. In Survey 2, participants complete 3 practice trials with clear examples. While no explicit feedback is given (to match the main experiment conditions), the examples demonstrate the range of the rating scale. Participants must select ratings for each practice trial, confirming they understand the response mechanism. In Survey 2, some trials will include clearly inappropriate headings paired with sentences (e.g., "Architecture" for a sentence about cooking pasta). Participants who consistently rate clearly inappropriate headings as highly appropriate (four or higher on a 7-point scale) may not understand the task. These responses will be flagged, and participants with poor attention check performance (> 30% failure rate) will be excluded from analyses of the pilot testing. Unusually fast or slow response times (statistical outliers) may indicate a lack of comprehension or engagement. The consent form and debriefing screen will provide contact information for the PI and Faculty Advisor. In the surveys, participants will be reminded of their right to withdraw at any time



without penalty. The ability to exercise this right serves as an additional indicator that participants understand the voluntary nature of their participation. Participants can simply close their browser window to withdraw. After completing all tasks, participants will view a debriefing screen that explains the purpose of the research in more detail, clarifies the rationale for the experimental design, provides contact information for follow-up questions, and thanks participants for their contribution. This debriefing serves as a final educational component and allows participants to confirm their understanding of what they participated in.

G.4.2 If written consent is not obtained, explain how researchers will document a participant's voluntarily agreement to participate in the research?

By clicking the "I agree to participate" link, participants confirm that they have read the consent form and decided that they will participate in the online surveys. All screening responses, consent records, and demographic data will be stored securely. Participants will be assigned anonymous Identity Codes, with no linking of Sona Systems identities to experimental data.

G.5 RESEARCH INVOLVING DECEPTION

G.5.1 Does the proposed research involve deception, e.g., through provision of misinformation, withholding information, etc.? ☐ Yes ☒ No

If **NO**, proceed to **Section H: Risks and Benefits to Subjects**

If **YES**, complete [Appendix L: Research Involving the Use of Deception](#)

SECTION H: RISKS AND BENEFITS TO SUBJECTS (BENEFICENCE)

H.1 RISK CLASSIFICATION

H.1.1 What is the overall risk classification for the proposed research?

NOTE: According to HHS regulations, minimal risk means, *"The probability and magnitude of harm or discomfort anticipated in the research are not greater in and of themselves than those ordinarily encountered in daily life or during the performance of routine physical or psychological examinations or tests"*.

☒ Minimal risk ☐ More than minimal risk

H.2 ANTICIPATED RISKS AND DISCOMFORTS

H.2.1 What are the possible anticipated risks and discomforts to the participants? Describe all possible risks including psychological, physical, sociological or economic harm (e.g., risk of criminal or civil liability, damage to their financial standing, employability, insurability, reputation, or stigmatization, etc.) Check all that apply and provide a description for each:



☒ **Physical Harms:** Medical research often involves exposure to minor pain, discomfort, or injury from invasive medical procedures, or harm from possible side effects of drugs and devices. Procedures commonly used in research may result in no more than minor discomfort (e.g., temporary dizziness, the pain associated with venipuncture). **Describe all possible physical harms that are anticipated during the research:**

This research involves minimal risk to participants; however, the following minor physical risks may be present: 1) participants may experience mild eye strain or visual fatigue similar to routine computer use for academic work, and 2) minor physical discomfort sitting at a computer for the study duration.

☒ **Psychological Harms:** Participation in research may result in undesired changes in thought processes and emotion (e.g., episodes of depression, confusion, feelings of stress, guilt, and loss of self-esteem). These changes may be transitory, recurrent, or permanent. Most psychological risks are minimal or transitory, but some research has the potential for causing serious psychological harm.

Stress and emotional discomfort such as feelings of guilt or embarrassment may arise simply for thinking or talking about one's own behavior or attitudes on sensitive topics such as drug use, sexual preferences, selfishness, and violence. These feelings may be aroused when the subject is being interviewed or filling out a questionnaire. Stress may also be induced when a researcher manipulates the subjects' environment – as when "emergencies" or fake "assaults" are staged to observe how passersby respond. The possibility of psychological harm can increase when behavioral research involves an element of deception. **Describe all possible psychological harms that might occur during the research:**

This research involves minimal risk to participants; however, the following minor psychological risks may be present: 1) the repetitive nature of making grammatical judgments and appropriateness ratings may cause minor boredom or frustration, similar to that experienced during standardized testing; 2) although the study is anonymous and participants are told there are "no right or wrong answers" for the rating task, some individuals may experience mild anxiety about their performance, similar to taking a quiz; 3) the tasks require sustained attention and decision-making over 45–55 minutes, which may result in mild mental fatigue similar to completing a homework assignment or reading assignment for class.

☒ **Privacy and Confidentiality:**

- In the research context invasion of privacy may involve either covert observation or "participant" observation of behavior that the subjects consider private.
- Can the invasion of privacy involved be justified considering the subjects' reasonable expectations of privacy in the situation under study?
- Are the potential benefits of the research of sufficient importance to justify the intrusion?
- Will the research involve the use of a subject's medical, school, or employment records? Access to such records for legitimate research purposes is generally acceptable, if the researcher protects the confidentiality of that information and obtains appropriate authorization from the individual.
- It is important to recognize that a breach of confidentiality may result in psychological harm to individuals (in the form of embarrassment, guilt, stress, and so forth).

Describe any possible situations that may be perceived as a risk to privacy and confidentiality:

Privacy and confidentiality risks: 1) participants will be made aware in the Consent for Online Surveys that "no guarantees can be made regarding the interception of data sent via the Internet by any third



party. Confidentiality will be maintained to the degree permitted by the technology used." These risks are substantially mitigated by the anonymous nature of data collection (no personally identifying information is linked to responses) and secure data storage procedures. While data collection is anonymous, participants recruited through the Psychology Research Participation system receive research credit for participation. There is a minimal risk that SONA participation records could theoretically be cross-referenced with study timing, though no identifying information is stored with experimental data.

☐ **Social and/or Economic Harms:** Some invasions of privacy and breaches of confidentiality may result in embarrassment within one's business or social group, loss of employment, or criminal prosecution. Areas of particular sensitivity are information regarding alcohol or drug abuse, mental illness, illegal activities, and sexual behavior. Confidentiality safeguards must be strong in these instances. Any anticipated costs to research participants should be described to prospective subjects during the consent process. **Describe any possible situations that may result in any social or economic harms:**

[CLICK HERE TO ENTER TEXT](#)

H.2.2 Describe any additional risks to participants or others:

N/A

H.2.3 How will risks be minimized and their potential impact to the participants or others be reduced?

- Does the research team have sufficient expertise and experience to conduct the research?
- Is the projected sample size sufficient to yield useful results?
- Is the data collected from standard-of-care procedures to avoid unnecessary risk, particularly for invasive procedures?
- Are adequate safeguards incorporated into the research design such as the presence of trained personnel and/or resources to respond to emergencies, and procedures to protect the confidentiality of the data (e.g., IT-approved databases for management and storage, adequate encryption, codes, and passwords).
- If appropriate, describe the plan for monitoring sensitive data.

The research design incorporates multiple safeguards to minimize risks and reduce their potential impact. Research Team Expertise and Qualifications: the Principal Investigator has completed CITI training in human subjects research protection, has received training in experimental design and research methodology through graduate coursework, and is conducting this research under close supervision with regular oversight. The Faculty Advisor has extensive experience supervising student research, is an expert in research methodologies in library and information science, and provides ongoing guidance on research design, ethical considerations, and data analysis. Consultation and Validation: experimental stimuli will be validated through consultation with DU librarians. The research design is adapted from the peer-reviewed methodology by Trott & Bergen (2023) in Psychological Review. Pilot testing with 20 participants recruited through Sona (with a minimum reporting threshold of 15 participants) is designed to collect diagnostic data prior to the main data collection period that will be part of a subsequent IRB application. Use of Established, Validated Procedures: the sensibility judgment task is a well-established psycholinguistic methodology used in dozens of published studies. The subject heading appropriateness rating paradigm is adapted from standard Likert-scale rating tasks commonly used in survey research. Stimuli are based on validated materials from Trott & Bergen



(2023). Online administration via PsychoPy and Pavlovia follows standard practices in experimental psychology. The study involves only reading, keyboard responses, and selecting ratings, i.e., entirely non-invasive behavioral observation. Participant Comfort and Well-being Measures: Practice trials allow participants to acclimate to tasks. A realistic time estimate will be provided in the consent form and SONA posting. Participants can withdraw at any time without penalty by simply closing their browser. This will be clearly stated in the consent form. Data Security and Confidentiality Protections: no personally identifying information is linked to experimental responses. Participants are assigned random numeric IDs by the online platform. No names, email addresses, IP addresses, or other identifiers are stored with experimental data. Sona Systems credit is granted separately without linking identity codes to experimental data. Data is hosted on Pavlovia (pavlovia.org), a platform specifically designed for secure behavioral research data collection. Pavlovia uses industry-standard encryption (SSL/TLS) for data transmission. Data is stored on password-protected, secure servers. Data access is restricted to the PI and the Faculty Advisor only. Downloaded experimental data will be stored and analyzed on password-protected computers. No data is shared with third parties. Data retention will follow DU policies. Only aggregate, anonymized data will be reported in publications. The study does not collect information about mental health, illegal activities, sexual behavior, substance use, or other sensitive topics. The demographic questions are minimal and not sensitive. Attention and Engagement Monitoring: attention check trials (e.g., clearly inappropriate headings) will identify disengaged or confused participants. Response time monitoring will detect outliers, suggesting inattention. Data from participants who fail attention checks will be excluded from analysis, ensuring only quality data are used. Technical Support and Troubleshooting: pilot testing will identify and resolve technical problems before the main data collection in a follow-on study. Compatibility testing will be done across multiple browsers (Chrome, Firefox, Safari). Clear technical requirements will be stated in the SONA posting (computer/laptop with keyboard required). Contact information for the PI is provided in the consent form and debriefing. Ethical Oversight and Monitoring: the study will not begin until full IRB approval or exemption is obtained. Any protocol modifications will be submitted to the IRB for approval before implementation. Ongoing Monitoring: the Faculty Advisor reviews study progress regularly. Weekly data quality checks will be conducted during the data collection period. Completion rates, attention check performance, and technical issues will be monitored continuously. The PI will pause recruitment if problems are identified. The debriefing form invites participants to contact the research team with concerns. Pilot testing of both experiments will collect qualitative feedback on task difficulty and clarity, appropriate session length, technical functionality, and any sources of confusion or discomfort. Procedures will be revised based on pilot feedback before the main data collection.

H.3 DIRECT BENEFITS

In order to justify the use of participants in a research study, the potential benefits of the study must be evaluated and weighed against the risks in the study. The benefit of a study can be to the participant in the study and/or the general community.

Therapeutic studies, such as behavioral interventions, are conducted with the intent to study alternative procedures for patient or client care. For a therapeutic study to be justified, the study procedure must offer care that is consistent with



other therapeutic options, there are multiple options for treatment, none of which are clearly preferred, and the risks are reasonable in relation to the potential benefit to participants.

Non-therapeutic studies seek to answer a scientific question without providing any treatment or direct benefit to a participant. For a non-therapeutic study to be approved, the risks in the study must be minimized according to sound scientific design, and the risks are reasonable in relation to the knowledge gained by the study. Although the participant may not directly benefit from the study the study may provide valuable information for a community or the general population.

NOTE: The DU IRB does not consider payment to be a benefit. Payments are used to encourage participation and should not be advertised as a benefit to participating in the study.

H.3.1. Describe the direct benefits to the research participants for participating in this research (i.e., to contribute to science, society, or to study alternative therapeutic methods)?

This is a non-therapeutic study designed to answer scientific questions about cognitive processing and knowledge organization systems. Participants may experience indirect benefits through their research participation, including advancing our understanding of how humans cognitively process word meanings and evaluate controlled vocabulary systems, as well as gaining firsthand experience with experimental research methodology. While participants do not receive direct benefits, this research has substantial potential to benefit society broadly through improvements to information retrieval systems and advancement of scientific knowledge. Library of Congress Subject Headings are used in thousands of libraries worldwide. Better-designed controlled vocabularies facilitate more efficient access to educational, research, and cultural resources, and more intuitive systems for users improve equity in information access. Researchers across all disciplines benefit from improved library discovery, and better information retrieval accelerates scientific and scholarly progress. Improvements to LCSH, which are also used in archives, museums, and special collections, benefit public access to cultural and historical resources and support discoverability of diverse traditions. This study advances the knowledge base of library and information science as a field and demonstrates the value of interdisciplinary research combining cognitive science and LIS. This study establishes a methodology for "cognitive warrant" that can be extended to other controlled vocabularies (e.g., MeSH or subject headings in other languages). These are, of course, potential rather than guaranteed benefits; however, even null findings would provide valuable evidence for the field, and the methodological contributions alone justify the research by establishing approaches for future studies.

SECTION I: CONFIDENTIALITY & PRIVACY

I.1 CONFIDENTIALITY

Confidentiality refers to the researcher's agreement with the participant about how the participant's identifiable private information will be handled, managed, and disseminated, including the research participants' authorization for the PI to view, share, and use their private information. Confidentiality means that the participants can be identified, but their identities are not revealed to anyone outside of the study; only the researcher knows the identities of the participants.



Anonymity means that there is no way for anyone (including the researcher) to personally identify participants in the study at any time. This means no personally identifying information can be collected in an anonymous study. Personally identifying information includes, but is not limited to, names, addresses, e-mail addresses, phone numbers, and IP addresses. Any study conducting face-to-face interaction, through in-person contact, using video technology (i.e., Zoom), or conducting any part of the study interventions via phone is NOT considered anonymous; this rules out virtually all qualitative research that involves collecting and analyzing non-numerical data (e.g., texts, video, or audio) to understand concepts, opinions, or experiences.

I.1.1 Will any demographic data or direct participant identifiers(referenced below)be collected or recorded, at any time, in your research? ☐ Yes ☒ No

If **YES**, indicate the type of data that will be recorded:

- ☐ Names/initials
- ☐ All geographic subdivisions smaller than a State, including street address, city, county, precinct, zip code, and their equivalent geocodes, except for the initial three digits of a zip code
- ☐ All elements of dates (except year) for dates directly related to an individual, including birth date, admission date, discharge date, date of death; and all ages over 89
- ☐ Telephone numbers
- ☐ Email addresses
- ☐ Social security numbers
- ☐ Medical record numbers
- ☐ Health plan beneficiary numbers
- ☐ Account numbers
- ☐ Certificate/license numbers
- ☐ Vehicle identifiers and serial numbers, including license plate numbers
- ☐ Web Universal Resource Locators (URLs)Other: [CLICK HERE TO ENTER TEXT](#)
- ☐ Internet Protocol (IP) address numbers
- ☐ Audio recordings and other biometric identifiers (fingerprints)
- ☐ Video recordings, full-face photographic images and any comparable images
- ☐ Any other unique identifying number, characteristic, or code (e.g. student ID numbers);
- ☐ Any information if used alone or in combination with other information could identify an individual who is a subject of the information

If **NO**, proceed to **section I.3 - Privacy**

I.1.2 Does the consent and other information presented to potential research participants adequately and clearly describe confidentiality risks? ☐ Yes ☐ No

[CLICK HERE TO ENTER TEXT](#)

I.1.3 Does the consent process and the informed consent document clearly delineate who will have access to the subject's information and under what circumstances data may be shared? ☐ Yes ☐ No

I.1.4 Will an individual study participant's data be shared with that participant? ☐ Yes ☐ No



If YES, are there circumstances or limitations on what data will be shared or how it will be shared with the participant? Please explain below.

[CLICK HERE TO ENTER TEXT](#)

I.2 CERTIFICATE OF CONFIDENTIALITY (CoC)

Certificates of Confidentiality (CoCs) protect the privacy of research subjects by prohibiting disclosure of identifiable, sensitive research information to anyone not connected to the research except when the subject consents or in a few other specific situations.

CoCs are issued automatically for any NIH-funded projects using identifiable, sensitive information. Receiving a CoC for research that is funded by other HHS-agencies, by non-HHS federal agencies, and for research supported by non-federal funders. To obtain a Certificate of Confidentiality, access [NIH Website: Certificates of Confidentiality](#).

I.2.1 If your project is NIH-funded, and involves identifiable, sensitive information, has information been included in the informed consent document? ☐ Yes ☐ No ☐ N/A

If no, refer to the [consent template for wording](#).

I.2.2 If your project is funded by non-HHS agencies and involves identifiable, sensitive information, will a CoC be obtained? ☐ Yes ☐ No

If **YES**, has CoC information been included in the informed consent document? ☐ Yes ☐ No

I.3 PRIVACY

Privacy is the control over the extent, timing, and circumstances of sharing oneself (physically, behaviorally, or intellectually) with others. For example, potential participants may not want to be seen entering a place or talking to an outsider (e.g., researcher) that may stigmatize them or increase potential risks by doing so. The evaluation of privacy also involves consideration of how the researcher accesses information from or about potential participants (e.g., recruitment process).

I.3.1 Describe the methods used to identify and obtain information about potential study participants who may meet the inclusion criteria for this study and maintain their privacy.

Potential participants will be identified exclusively through the University of Denver Psychology Department's Research Participation System. This system is specifically designed to protect participant privacy during the recruitment process. Participants browse available studies on their own SONA accounts and choose which studies to view. No potential participant is contacted or identified unless they voluntarily express interest. Even after sign-up, the online nature of this study means no personally identifying information is linked to experimental data. Eligibility screening occurs before the online surveys. No personally identifiable information is collected during screening. The PI only knows aggregate information, cannot identify specific individuals who meet inclusion criteria, and cannot target or exclude specific individuals. Interested students receive a link from the Research Participation System (not directly from the PI) and complete the online survey anonymously. The PI grants credit through SONA without knowing which specific person completed the study.



I.3.2 Describe the methods and process to contact and recruit potential participants who meet the inclusion criteria while ensuring their privacy is respected.

Recruitment occurs entirely through the SONA platform using a self-selection model that maximizes privacy protection. Step 1: The PI creates the study postings in the Research Participation System. The researcher does not directly contact anyone. Students discover the surveys on their own when browsing available opportunities. Step 2: Students who are interested voluntarily sign up through their SONA account. Sign-up requires no additional information beyond what's already in SONA (which is not shared with the researcher). Participation is entirely voluntary and self-initiated. No one is contacted, solicited, or pressured. Step 3: SONA automatically sends the study link to participants who pass the screening and sign the consent form. Each participant receives a unique anonymous link for credit tracking purposes only. There is no direct communication from the researcher to participant. Step 4: Participant clicks the link and completes the survey online at the time/location of their choosing. No identifying information is collected in the experimental data. Step 5: Upon survey completion, Pavlovia sends a completion signal to SONA. The PI grants research credit through the SONA interface, and the credit appears on the student's SONA account. The researcher cannot link specific experimental data to specific users.

I.3.3 Describe the settings in which an individual will be interacting with the investigator or research team member that provides adequate privacy.

Note: If any study interaction will occur remotely, describe what information or safeguards will be implemented to ensure the study participant's privacy is protected during the remote study interaction.

This study is conducted 100% online with no in-person interaction between researchers and participants. All study activities occur remotely via the participant's personal computer. Participants choose where to complete the study; there is no need to travel to a research lab or meet a researcher in person. Participants choose when to participate within their schedule. Participants can use a private/incognito browser mode if desired. This study does not activate or require a camera or microphone. This study does not require or use screen-sharing or recording technology. This study involves only text display and keyboard/mouse responses. The only recorded data are response choices, response times, and minimal demographics. The study design ensures privacy even if someone else is present: there is no on-screen display of "participant ID" or identifying information, rating scales do not display the participant's previous responses, and the content is not controversial, embarrassing, or sensitive. Participants can withdraw simply by closing the browser window. There is no need to contact a researcher to withdraw, and no explanation is required. The withdrawal is completely private; the researcher doesn't know who withdrew. The study does not store IP addresses, collect location data, or collect browser type, operating system, or device identifiers. Only essential functional cookies (no tracking or advertising cookies) are used. Each participant is assigned an anonymous numeric ID unrelated to their identity. Only the Sona System knows who participated in the study (for granting of credit). The Pavlovia platform and researcher only know what responses were given (experimental data). These two pieces of information are never linked; no dataset contains both identity and responses. All data transmitted between the participant's computer and Pavlovia servers uses industry-standard encryption (<https://>). The PI and Faculty Advisor are the only personnel authorized to have access to the data. When the researcher downloads experimental data, a secure, encrypted connection is used, and the downloaded data will be stored on password-protected devices. Any cloud backup will be on the PI's institutional Microsoft OneDrive



account. Each session is independent. Each participant's involvement ends upon survey submission; there is no ongoing relationship that could compromise privacy.

I.3.4 If information will be obtained about individuals other than the “target participants,” (e.g., a subject provides information about a family member for a survey), describe the process for obtaining this individual’s information and how their privacy will be protected and respected. If not applicable to this study, enter “N/A”.
N/A

SECTION J: RESEARCH DATA MANAGEMENT

The retention of recorded and retrievable research data is of utmost importance for the progress of scientific integrity. The investigator is responsible for accurately recording, retaining, and storing research data as well as ensuring the rights and privacy of the research participants and their identifiable information is protected. Research records should include sufficient detail to permit examination for the purpose replicating the research, responding to questions that may result from unintentional error or misinterpretation, establishing authenticity of the records, and confirming the validity of the conclusions as well as maintained to preserve the confidentiality of a participant’s private data.

J.1 IDENTIFIERS AND CODING

Research involving the initial collection of private data from living individuals will always require IRB approval whether or not the data are “coded” or “de-identified” subsequently. Researchers who will use coded or de-identified private information or specimens must provide adequate information about how the proposed research data will be or was coded or de-identified. As an investigator who proposes to obtain and use “de-identified data”, certain conditions must be met to ensure that the identities of research subjects cannot be ascertained with the data. If the conditions for de-identification are not met and the data may simply be “coded”, **and** the key to the code exists somewhere, additional information must be provided to describe how the coded or de-identified private information or biospecimens will be secured. To further define the terms used with identifying and coding data, please refer to the following terms and definitions:

A direct identifier is a specific element of data, such as a name, social security number, or address, which directly connects a participant and their data.

An indirect identifier is any element or condition of data which potentially allows someone to connect a research participant and their data.

Coded Data: Coded data has identifying information (such as name or social security number) that would enable a research team to ascertain the identity of the individual to whom the private information or specimens pertain and is replaced with a “code” (number, letter, symbol, or combination). **Note that simply coding the data does not make that data anonymous or de-identified.** In most studies involving coded data, a key to decipher the code typically exists, and the key enables linkage of the private information or specimens to individuals.

Anonymous Data: Data are anonymous if no one, not even the researcher, can connect the data to the individual who provided it. If no identifying information is ever collected from individuals (including direct identifiers as listed in [section I.1.1](#)), the data might be anonymous. However, researchers should be aware that collection of indirect identifiers, (e.g., information regarding other unique individual characteristics) might make it possible to identify an individual from a



pool of subjects. For example, a study participant whose data may identify them as a member of a minority ethnic group and a particular sex might be identifiable from even a large data pool.

Data De-identified: Data are considered de-identified when all direct or indirect identifiers or codes linking the data to the individual subject's identity are destroyed. This means that any of the 18 identifiers listing in **section I.1.5**, of the individual or of relatives, employers, or household members of the individual are removed.

J.1.1 Will this study collect or obtain any identifiable data about the study participants? ☐ Yes ☒ No

If **NO**, proceed to section J.2 – **Research Data Collection**

If **YES**, continue to **section J.1.2**

J.1.2 Describe the data security measures that will be implemented to protect the identity of and/or confidential information obtained about individuals who participate as subjects in research. Explain why it is necessary to maintain such identifiers and describe the coding system you will use to protect the subject's privacy against disclosure.

[CLICK HERE TO ENTER TEXT](#)

J.2 RESEARCH DATA COLLECTION

J.2.1 Specify how information will be obtained? (check all that apply)

- ☒ **Information is publicly available** (e.g., Public Use Database) Specify source: Data and code to reproduce the original, preregistered analyses of Trott & Bergen (2023) are available on Open Science Framework and GitHub.
- ☒ **De-identified data** is collected by the investigator, with no direct or indirect identifiers and all data is anonymous
- ☐ **De-identified data** is obtained from an outside entity. Specify source: [CLICK HERE TO ENTER TEXT](#)
- ☐ **De-identified data** obtained from a DU-affiliated investigator, without codes or any direct or indirect identifiers. Specify DU-affiliated investigator and IRBNet Project number (if applicable): [CLICK HERE TO ENTER TEXT](#)
- ☐ **Identifiable data** collected and coded as part of the study, with the key to the codes maintained only by the investigator.
- ☐ **Identifiable data** collected and then de-identified by the study investigator, with no codes or any direct or indirect identifiers,
- ☐ **Identifiable data** collected and then de-identified (all identifiers stripped) by an independent individual not associated with the study, with no codes or any direct or indirect identifiers (e.g., obtaining student data from a DU department or client data from a DU clinic). Specify the responsible individual and the source of data: [CLICK HERE TO ENTER TEXT](#)
- ☐ **Storage or maintenance of existing or prospectively collected identifiable private information or identifiable biospecimens** obtained through an outside entity. Specify source: [CLICK HERE TO ENTER TEXT](#)
Data Use Agreement (DUA) or Materials Transfer Agreement (MTA) may be required
- ☐ **Secondary research using identifiable private information or identifiable biospecimens** will be conducted. Provide source of data: [CLICK HERE TO ENTER TEXT](#)
Data Use Agreement (DUA) or Materials Transfer Agreement (MTA) may be required



- ☐ Data collected with a crowdsourcing platform (MTurk, Prolific, etc.)

J.3 DATA MANAGEMENT

Ownership of Data

J.3.1 If your research plans include the collection or study of existing data, pathological or diagnostic specimens, please specify whether data will be restricted and if an agreement is in place prohibiting release of private identifiable information or if you will be provided the key to coded information and provide such documentation.

If not applicable, please indicate "N/A".

N/A

Retaining and Securing Data

J.3.2 How will the research data and/or specimens be protected against inappropriate use or disclosure?

Confidentiality is best ensured through proper data management and security. It is strongly recommended that all researchers consult with DU IT to confirm that their proposed data security system meets the appropriate requirements to protect the research data.

(Check ALL that apply)

- ☐ Locked office on campus
- ☐ Locked storage unit on campus
- ☒ Restricted access to authorized study personnel
- ☐ Secure computer/laptop
- ☒ Individual ID plus password protection
- ☐ Encryption of digital data
- ☐ Network Restrictions
- ☒ Security software (firewall, antivirus, anti-intrusion) is installed and regularly updated in all servers, workstations, laptops, and other devices used in the study
- ☐ Restrictions on copying study related materials
- ☐ Destruction of source data immediately after data collection (to preserve anonymity of participants)
- ☐ Audio and/or video recordings will be transcribed and then will be destroyed
- ☐ Audio and/or video recordings will be modified to eliminate the possibility that study participants could be identified
- ☐ Photos or images will be modified to eliminate the possibility that study participants could be identified
- ☐ Study personnel will sign statements agreeing to protect security and confidentiality of study information
- ☐ Access rights are terminated when authorized study personnel leave the study
- ☐ Not Applicable
- ☐ Other: [CLICK HERE TO ENTER TEXT](#)

Storage of Data

J.3.3 If a secure DU data storage platform will be utilized, identify the system(s) below.

- ☒ OneDrive
- ☐ RedCAP
- ☐ Google Drive
- ☐ Other: [CLICK HERE TO ENTER TEXT](#)



☐ N/A

J.3.4 What will happen to the research data and/or specimens at the conclusion of the study? Include paper records, audio and video recordings, field notes, transcriptions, etc. in your response. If study materials are to be retained for at least three years after at the conclusion of the research, describe how secure storage of those materials will be maintained.

- ☐ Direct identifiers and/or the key to the codes will be destroyed upon completion of the research (all data/specimens will be stripped of identifying information and/or the key to codes destroyed, paper documents shredded, electronic files purged, electronic media securely erased)
- ☐ Retain for study recordkeeping purposes per institutional policy
- ☐ Retained by the investigator for future research use (submit Appendix I).
- ☐ Retained to create data or tissue repository/bank (submit Appendix I).
- ☐ Restricted use of data will be destroyed or returned to the source.
- ☒ No direct or indirect identifiers are being collected. The anonymous data and/or specimens will be retained at the discretion of the investigator.
- ☐ This research is a clinical trial conducted under FDA regulations. Direct identifiers and/or the key to the codes will be destroyed as directed by the sponsor in accordance with FDA regulations.
- ☐ Other [CLICK HERE TO ENTER TEXT](#)

NOTE: Per DU IRB Policy, documentation of the signed informed consent of the subjects, written protocol, and other records related to conducting research that are typically held by investigators must be retained for at least three years after completion of the research.

J.3.5 If identifiable study data, direct or indirect identifiers, are stored on a networked computer or device, either on campus or off-campus, transmitted over a network, and/or stored on a removable medium, what security measures will be implemented AFTER the research has been completed. If not applicable, please indicate "N/A".

N/A

Transfer of Data

J.3.6 In the event that outside organizations are involved (in data gathering, processing, and storage), what safeguards or procedures will be incorporated to ensure the rights of the subjects will be guaranteed by that agency? If not applicable, please indicate "N/A".

N/A

SECTION K: DATA USE AGREEMENTS/MATERIAL TRANSFER AGREEMENTS

K.1 DATA USE AGREEMENTS

A data use agreement (DUA) is a written contract governing the transfer of data, including human subject data, between entities for the purposes of research. These agreements can be set up between academic institutions, non-profits, government agencies, and/or corporate entities. The Data Agreement will govern data ownership, permitted uses of the data, publication of results, development of inventions, disposal of the data, and liability.



If **de-identified data** is requested from an outside entity, without any codes or access to the key to the codes, it is at the discretion of the outside entity whether they will require a DUA.

K.1.1 Will this research involve RECEIVING identifiable data from or TRANSFERRING identifiable data to an outside entity? ☐ Yes ☒ No

If **NO**, proceed to **K.2**.

If **YES**, complete the Office of Intellectual Property and Technology Transfer (OIPTT) intake form [here](#). The PI is responsible for attaching the fully executed DUA document to their approved IRBNet project as a new package labeled "Other."

NOTE: If the PI is a student, the DUA must be submitted by the Faculty Sponsor.

K.2 MATERIAL TRANSFER AGREEMENTS

A material transfer agreement is a contract governing the transfer of tangible research materials (biological materials) between institutions for the purposes of research.

K.2.1 Will this research involve RECEIVING or TRANSFERRING tangible research materials? ☐ Yes ☒ No

If **YES**, complete the Office of Intellectual Property and Technology Transfer (OIPTT) intake form [here](#).

SECTION L: MULTI-SITE/ COLLABORATIVE RESEARCH WITH EXTERNAL INSTITUTION OR ORGANIZATIONS & UNAFFILIATED INVESTIGATORS

If you are not collaborating with another institution or organization, please leave Section L blank. Proceed to Final Instructions on the last page.

L.1.1. Are you requesting that DU IRB serve as the IRB of record for this study? ☐ Yes ☐ No

If **YES**, is this external/unaffiliated individual(s) associated with another institution or organization that has a Federalwide Assurance (FWA)? ☐ Yes ☐ No ☐ Unsure: If unsure, contact the IRB Office.

If **YES**, complete and attach [Appendix P: Request the DU IRB to Serve as the IRB of Record](#) to your IRBNet package submission

If **NO**, **DU is unable to execute a reliance agreement with another institution or organization that does not maintain a Federalwide Assurance (FWA). Proceed to L.2.**

NOTE: The DU IRB will evaluate the ceding request and make a determination whether a reliance agreement can be executed between DU and the other institution or organization.



L.1.2 Is the external organization or institution that requests a ceding arrangement with DU an international entity?

☐ Yes ☐ No

If **YES**, **no ceding arrangement may be executed between DU and a foreign country** and each investigator must obtain their own IRB or research ethics committee review.

If **NO**, proceed to **L.1.3**

L.1.3 Does the external/unaffiliated individual(s) institution or organization have an internal Institutional Review Board (IRB)? ☐ Yes ☐ No

If **YES**, the external/unaffiliated individual(s) must obtain approval from their own institution's IRB to allow a formal ceding arrangement to occur between DU and the unaffiliated individual's own institution.

- If a ceding arrangement is requested to have DU serve as the IRB of Record, complete and attach [Appendix P: Request the DU IRB to Serve as the IRB of Record](#) to your IRBNet package submission

If **NO**, the external/unaffiliated individual(s) must be listed in **Section A.3 - Table 3.1**. The IRB will need to evaluate whether the individual(s) are qualified to be added as an unaffiliated investigator on the research project. If approved, each external individual must complete the mandatory human subjects protection training for unaffiliated investigators through the DU CITI Program. **COMPLETE SECTION L.2**

L.2 ADDING AN UNAFFILIATED INDEPENDENT INVESTIGATOR OR INDIVIDUAL

To add an unaffiliated independent investigator/individual who is **not eligible for ceding**, the [Individual Investigator Agreement \(IIA\)](#) and the [APPENDIX R: Adding an Unaffiliated Independent Individual](#) must be completed and attached to this IRBNet package for review. All individuals listed in Section A.3, [Table 3.1 – Unaffiliated Investigators/Individuals](#) must complete the mandatory DU human subjects protection training curriculum created for unaffiliated investigators offered through the DU CITI Program.

L.3 COLLABORATIVE RESEARCH WITH AN EXTERNAL ORGANIZATION/SCHOOL/INSTITUTION

L.3..1 Will any recruitment or research interventions occur at any off-campus site or sites? ☐ Yes ☐ No

If **YES**, provide a complete list of the external organizations, schools, or institutions that will be involved in the recruitment or participating in any of the research interventions and attach a letter of support as part of your IRBNet package.

[CLICK HERE TO ENTER TEXT](#)

If research is conducted at an off campus site (i.e., community center, school, etc.) or is recruiting potential study participants from that site (i.e., employees, students, etc.), a signed **letter of support** must be obtained from the external site confirming their support of the proposed research and their approval to have the research be conducted at their site. Post the letters of support from all off-campus sites in the IRBNet project submission. Letters of support must be obtained and reviewed before final IRB approval will be issued.



FINAL INSTRUCTIONS

Before submitting your IRBNet Application, please make sure that the PI has electronically signed the package. If the PI is a student, then the PI and Faculty Sponsor must electronically sign the package.

Research staff may not sign on behalf of the PI or Faculty Sponsor.

Questions? Contact IRBAdmin@du.edu.